

**99 candidate cardiovascular RCTs
(landmark trials; prior emulation
initiatives)**

Excluded, 61 [counts by reason to be filled from the screening log]

- Placebo-only design without an accepted active proxy, n = []
- Primary endpoint not ascertainable from EHR data, n = []
- Treatment strategy not identifiable, n = []
- Insufficient size (<300 with ECG in smaller arm or <50 events), n = []
- Near-duplicate cohort (>80% shared records), n = []
- Not screened / previously analysed design variant, n = []

38 trials emulated
AF 12, diabetes 9, HF 5,
hypertension 5, ACS 2, other 5

Excluded after emulation-quality grading, 6 (limited, ≥ 4 points)
ALLHAT, ASCOT-BPLA, DECLARE-TIMI 58,
ONTARGET, REWIND, VALUE

32 trials analysed (primary set)
3 excellent, 12 good, 17 moderate
AF 12, diabetes 7, HF 5,
hypertension 2, ACS 2, other 4

External validation (MIMIC-IV): 7 cardiovascular trials meeting the
same feasibility criteria + PEPTIC (negative control)
UK Biobank: cardiac-MRI plasmode